Lecture 13: Why Are People Different?: Differences (full transcript)

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Overview: Why are people different from one another? This lecture addresses this question by reviewing the latest theories and research in psychology on two traits in particular: personality and intelligence. Students will hear about how these traits are measured, why they may differ across individuals and groups, and whether they are influenced at all by one's genes, parents or environment.

Chapter 1. Personality, Intelligence and Determining Difference [00:00:00]

Professor Paul Bloom: What we've been talking about so far in the course are human universals, what everybody shares. So, we've been talking about language, about rationality, about perception, about the emotions, about universals of development, and we've been talking about what people share. But honestly, what a lot of us are very interested in is why we're different and the nature of these differences and the explanation for them. And that's what we'll turn to today.

So first, we'll discuss how are people different, different theories about what makes you different in a psychological way from the person sitting next to you, and then we'll review different theories about why people are different. And this is the class which is going to bother the most people. It's not dualism. It's not evolution. It's this because the scientific findings on human psychological differences are, to many of us, shocking and unbelievable. And I will just try to persuade you to take them seriously.

Okay. So, how are people different? Well, there's all sorts of ways. Your sexual identity — It is at the core of your being for almost all of us whether you're male or female. How we refer to you in language, what pronoun we use, is indexed on how we — on your — on how — whether you're male or female and related to that though imperfectly is your sexual orientation, who you're attracted to. The question of why some of us think of ourselves as males and others as females, and the question of why some of us would ideally want to have sex with males, others with females, others with both, and then a few others who have harder to define desires, is such a good question that we're going to talk about it after spring break while all the sexual desire has been spent and you could focus on [laughter] on a scientific discussion of this — not that I recommend you do that on spring break.

How happy are you? This is also such a good topic it's going to get its own class. The very last class of the semester is devoted to happiness and the question of what makes people happy, what makes people unhappy, and what makes people differ in their happiness. If I asked you to rank how happy you are from a scale of 1 to 10, the numbers would differ across this room. And there's different theories as to why. Your success and failure in life — This is somewhat interesting because you could study this in more or less objective ways. We don't have to ask people. We could look at your relationships, how they begin, how they end, your job satisfaction. We could look at your criminal records. Some of you are going to see time. Most will not. Some of you will get into little troubles all through your life. Some of

you already have seen the inside of a police station, possibly a lineup. Others couldn't go near such a thing. What determines that?

And at the root of all human differences are two main factors. And so, I want to talk about the two main interesting factors. One is personality. The other is intelligence. And this is what — These are the differences I'll talk about today first from the standpoint of how do we characterize them, how do we explain them, and then from the standpoint of why these differences exist in the first place.

One way to characterize personality is in terms of people's style with dealing with — in dealing with the world and particularly their style with dealing — in dealing with other people. So, you take a simple character you know of and you could talk about that person's personality. You could talk about it in terms of being impulsive, irresponsible, sometimes lazy, good-hearted. You could compare that person's personality with other people's personalities such as my colleague who gave a talk last class. He's wonderful. He's responsible and reliable and very kind [laughter] and different from Homer. And so, this difference is a difference in personality.

Now, when we talk about personality we're talking about something else as well. We're talking about a stable trait across situations and time. So, if all of a sudden the person next to you kind of smacks you in the head, you might be angry but we wouldn't call that "personality" because that's something that's a result of a situation. We'd all feel that way in that situation. It's "personality" if you walk around all the time angry. That'd be a stable trait. That'd be something you carry around with you and that's what we mean by personality.

Now, how do we scientifically characterize differences in personality? And it's a deep question. There's been a lot of attempts to do so. Any assessment has — Any good assessment has to satisfy two conditions. And these are terms which are going to show up all over psychological research but it's particularly relevant for this sort of measure. One is "reliability." Reliability means there is not measurement error. And one crude way to think about reliability is, a test is reliable if you test the same person at different times and you get the same result. My bathroom scale is reliable if whenever I stand on it, it gives me more or less the same number. It's not reliable if it's off by ten pounds in the course of a day. Similarly, if I give you a personality test now and it says that you're anxious and defensive, well — and then give it to you tomorrow and it says you're calm and open minded, it's not a reliable test. So, reliable is something you could trust over time.

"Validity" is something different. Validity is that your test measures what it's supposed to measure. So, validity means it's sort of a good test. Forget about how reliable it is. Does it tap what you're interested in? So, for example, suppose I determine your intelligence by the date of your birth. I figure out what day you were born and I have a theory that, from that, predicts how smart you are. That's my intelligence test, the date of your birth. Maybe people born in January are the dumbest, people born in December are the smartest. Is that — I was born on Christmas Eve. [laughter] Is that a reliable test? Yes, it's a wonderfully reliable test. I'll test you today; I'll test you tomorrow; I'll test you next year; I'll test you the day you die; I'll get the same IQ score. Is it a valid test? It's a joke. It's absolutely not a valid test. It has nothing to do with intelligence. But you noticed these are two different things. Something can be reliable but not valid and something can be valid and not reliable.

Now, there are no shortage of personality tests. You could get them all over the place including on the web. So, I took one recently. I took "which super hero are you?" [laughter] And it's a series of questions determining what super hero you are. You could take this yourself if you want to. The same web page, by the way, offers you a test in whether you're "hot" or not. We'll discuss that later. And when I did this [laughter] it told me I was Batman [laughter] and "you are dark, love gadgets, and have vowed to help the innocent not suffer the pain you have endured." Now, the honest — [laughter] Now, to be honest though, it's neither reliable nor valid. When I first did the test I came up as "The Incredible Hulk." I then changed my answers a bit and was "Wonder Woman." [laughter] And finally, out of frustration, I carefully tailored my answers so I would be Batman. But the fact that I can do that, well, raises questions about both the reliability of this measure and its validity.

Here is an example – a real world example. This is, in black and white form, a version of the Rorschach test, the Rorschach inkblot test. How many people have heard of the Rorschach test? Okay. Is there anybody here who has actually, in any sort of situation, taken a Rorschach test? Some people scattered in the room have taken them. It was originally used only for psychiatric cases but then became extremely common. About eighty percent of clinical psychologists claim to use it and most graduate programs in the American Psychological Association who are accredited teach it. Catholic seminaries use it for people who want to join the seminary.

It was invented by a guy named Herman Rorschach. He devoted his entire life to the inkblot test. His nickname when he was a teenager – I am not kidding you – was "Inkblot." [laughter] And the idea is by looking at these inkblots and then seeing what somebody says you get great insights into the nature of their personality, into what they are. Anybody want to try it? Come on. Yes. What do you see?

Student: I see two people holding hands pressed together.

Professor Paul Bloom: Two people holding hands pressed together. Very good. Anybody have a different reading? Yes, in back. Yes. Yes.

Student: Dancing bears.

Professor Paul Bloom: Dancing bears. Okay. Good.[laughter] Good. Okay. I got to write your name down — [laughs][laughter] report you to health — No. Dancing bears, very good. Anybody else? One other. Yes.

Student: A man in a ski mask.

Professor Paul Bloom: A man in a ski mask. Well, it turns out that there are right answers and wrong answers to the Rorschach test. According to the test, and this is from a real Rorschach test, "it is important to see the blot as two human figures, usually females or clowns." Good work over there. "If you don't, it's seen as a sign you have problems relating to people." [laughter] If you want to go for "a cave entrance" or "butterfly" or "vagina," that's also okay. [laughter]

Now, the Rorschach test is transcendently useless. It has been studied and explored and it is as useless as throwing dice. It is as useless as tea leaves. Nonetheless, people love it and it's used all over the place. It is used for example in child custody cases. If you have broken up with your partner and you guys are quarreling over who gets to keep the kids, you might find yourself in a shrink's office looking at this. And in fact, this is why they end up on the web. There are services. There are people who have

been kind enough to put on the web these inkblots, including the right answers to them. But they are worthless as psychological measures.

Can we do better? Well, we probably can. Gordon Allport did a study where he went through the dictionary and took all of the traits that he believed to be related to personality and he got eighteen thousand of them. But what was interesting was they weren't necessarily independent traits. So, the traits like "friendly, sociable, welcoming, warm-hearted" seemed to all tap the same thing. So, Cattell and many others tried to narrow it down, tried to ask the question, "In how many ways are people's personalities different from one another?" How many parameters of difference do you need? How many numbers can I give you that would narrow you in and say what personality you are?

One approach was from Eysenck, who claimed there were just two. You could be somewhere on the scale of introverted-extroverted, and somewhere on the scale of neurotic and stable. And since there's basically two types of traits with two settings for each, there are basically four types of people. Later on he added another trait which he described as "psychoticism versus non-psychoticism" that crudely meant whether you're aggressive or empathetic. And then you have three traits with two settings each giving you eight types of people. Later on Cattell dropped it down into sixteen factors. So, these sixteen personality factors are sixteen ways people would differ. And so, if I asked you to describe your roommate along these sixteen dimensions, you should be able to do so.

Chapter 2. Measuring Personality with "The Big Five" [00:13:53]

More recently, people have come to the conclusion that two or three is too few, but sixteen might be too many. And there's a psychological consensus on what's been known as "The Big Five." And "The Big Five" personality factors are these, and what this means is when we talk about each other and use adjectives, the claim is we could do so in thousands of different ways, but deep down we're talking about one of these five dimensions. This means that when a psychological test measures something about somebody, about their personality, if it's a good test it's measuring one of these five things. And it means that, as people interacting with one another in the world, these are the five things that we're interested in. So, one of them is "neurotic versus stable." Is somebody sort of nutty and worrying or are they calm? "Extrovert versus introvert." "Open to experience versus closed to experience." "Agreeable," which is courteous, friendly versus non agreeable, rude, selfish. And "conscientious versus not conscientious," careful versus careless, reliable versus undependable. A good way to think about these things is in terms of the word "ocean," o-c-e-a-n. The first letter captures openness, conscientiousness, extroversion, agreeableness, and neuroticism. And the claim is those are the four — the five fundamental ways in which people differ from one another.

Well, why should we believe this? Why should we take this theory seriously? Well, there's actually some evidence for it. It seems to have some reliability in that it's stable over time. So, if you test people over years — If I test your personality now on the five traits and test you five years from now, it will not have changed much. And once you pass the age of thirty, it's very stable indeed. If you think about your parents and then give Mom and Dad a mental test on where they stand on each of the five traits, ten years from now Mom and Dad will still be there. It also seems to get agreement across multiple observers. So, if I ask for each of their five traits — If I ask your roommate what he or she thinks of you, then I ask your professor what he or she thinks of you and your mom what he or — what she thinks of

you, [laughter] how would — back to gender — How would they match up? They tend to overlap a lot. You walk around and you leave — and your personality leaves a trail in the minds of people around you. And this trail is characterized in terms of these five dimensions.

Finally, it seems to be — predict real-world behavior. If this didn't have anything to do with the real world, you wouldn't be very happy calling it valid, you wouldn't take it seriously as a test, but it does. So, conscientiousness — how you score on a conscientious scale, relates to how faithful you are to your spouse. How openness — open you are on a psychological personality test relates to how likely you are to change your job. "Extroverts" look people in the eye more and have more sexual partners because they're extroverts. So, these are real scales. The "Batman, Hulk, Wonder Woman" doesn't correspond to anything in the real world, but where you stand on each of these five dimensions does seem to capture it.

As an example of the agreement, by the way, somebody did a study of several of the characters on the television show "The Simpsons" because they wanted to find characters which everybody knew. And they had thirteen subjects judge these Simpson characters on each of the five dimensions. These is "openness, neuroticism, conscientiousness, and extroversion" and they found considerable agreement. And this isn't actually — What I've covered up [on the slide] is the "agreeableness." So, for those of you who have never seen the television show, this is all going to be confusing, but those of you who have, can you guess which characters would be particularly agreeable? Anybody guess. Yeah.

Student: Flanders

Professor Paul Bloom: Flanders. You are right. The most agreeable people are Flanders and Marge. Who would not so agreeable?

Student: Krusty

Professor Paul Bloom: Krusty is actually — Krusty is a complicated case [laughter] but Mr. Burns — but also — Where is he? Oh, he's not — Nelson, where's Nelson? Anyway, there's Nelson. You get strong consensus that Ned Flanders and Marge Simpson are highly agreeable people, 6.27 and 5.46, while Mr. Burns and Nelson are very low. Nelson's the little kid that when trouble happens he goes, "Ha ha." And that's a psychological sign for low agreeability. [laughter]

Okay. That's all I want to say at this point about personality and how we measure it and, again, we're going to get back to it later when we talk about differences in personality.

Chapter 3. Defining and Measuring Intelligence [00:19:47]

Now, I want to deal with the second big difference. The second big difference is intelligence. Now, how do you define intelligence? There's no easy definition. Like personality, it's kind of difficult to get your fingers on what we're talking about here. In one survey they asked 1,000 experts to define intelligence. And some answers showed up over and over again. So, just about everybody said intelligence involves abstract reasoning, problem solving, and the capacity to acquire knowledge. That's at the core of being smart. Other people mentioned things like memory, mental speed, language, math, mental speed again, knowledge, and creativity also as hallmarks for intelligence. And again, it might be difficult to define it but you have a gut feeling about what it is.

So, you know Homer is actually — and this is part of the show — is actually of limited intelligence. My colleague is of very high intelligence, a wonderful fellow, [laughter] but he's probably not as smart as that guy [pointing to a picture of Einstein] who is really, really smart. And this guy, Ralph Wiggum, is particularly stupid. [laughter] And so you have a range. And it's important to figure out how to characterize it; this is what research does, but there's a gut feeling that there are some people who are smart and other people who are very smart and some people who are dumb and others who are very dumb. What you want to do, from a scientific standpoint, is characterize this in a more robust and interesting fashion. And the textbook has a nice review of the history of attempts to define and measure intelligence, but there is a couple of ideas I want to focus on.

One is an idea developed by Spearman, which is there's two types of intelligence. There is "G" and there is "S." "S" is your ability on specific tests. So, if there is ten tests that you're given as part of an IQ test, ten subtests, you'll get a different score on each of the subtests. There'll be a math test and a reading test and a spatial test and you'll get different scores. "G" refers to a general intelligence. And the general intelligence is something you bring to each of the tests in common. So, this is diagrammed here. You have these six tests. For each of them there is an "S" and then above that there is a "G."

Now, "G" is a very important notion. The term "G" is used by psychologists a lot even in casual conversation. People say, "So, what do you think of him?" "I think he is high 'G.'" And what you mean is he's a smart guy. Why do you need "G?" Well, you wouldn't need "G" if your performance on each of these tests had nothing to do with each other. If the tests were genuinely separate, there'd be no general intelligence. But what people find over and over again is that when it comes to explaining people's performance on multiple intellectual tasks, there's two things going on. There's how good there is — they are on the specific task, but then there's also a sort of general correlation that people bring to the tasks.

And I could express this with an athletic analogy. Imagine I'm running a gym and we have all of these different athletic tests. So, we have a running test, we have a basketball shooting test, a swimming test, fencing, a list of ten of them. Now, each of you go through each of the tests and then you'll each get ten scores. But what we'll discover is that the scores are not independent of one another. People who are good at one athletic thing tend to be good at another. If there's somebody who's really good at running and swimming, odds are they're probably pretty good at climbing. And the same thing holds for IQ, which is above and beyond how good people are at specific things there seems to be a factor as to how well they are in general. And this factor is known as "G."

Now, there's, again, an extensive history of modern intelligence tests and what's really interesting is the tests now. What you need to know about the modern tests, the Wechsler test for both adults and children, is how they're scored. The way they are scored is that 100 is average. So, it's just automatic. Whatever the average is is 100. It's as if I did the Midterm — graded the Midterm, computed the average, gave everybody who got the average 100, said your score is 100. It's just the average. It works on the normal curve and what this means is that it works so that the majority, 68%, get between 85 and 115 on their IQ test. The vast majority, 95%, get between 70 and 130. If you are, say, above 145 IQ, which I imagine some people in the room are, you belong to 0.13% of the population. That's the way IQ tests work.

Now, this is about IQ tests. We could now ask about their reliability and their validity. What do they mean? Well, this has turned out to be a matter of extreme debate. This [slide] just reiterates what I just said. A lot of the debate was spawned by the book by Herrnstein and Murray about — called The Bell Curve. And in The Bell Curve these authors made the argument that IQ matters immensely for everyday life and that people's status in society – how rich they are and how successful they are – follows from their IQ as measured in standard IQ tests. Now, this book made a lot of claims and it's probably before many of you — many of your time, but spawned huge controversy. And as a result of this controversy some interesting papers came out.

One response to the Herrnstein and Murray book was by the American Psychological Association, which put together a group of fifty leading researchers in intelligence to write a report on what they thought about intelligence — what they thought about, "Does IQ matter? How does IQ relate to intelligence? How does — what's the different — why are people different in intelligence? Why do different human groups differ in intelligence?" and so on. At the same time, there was also another group of IQ researchers, not quite the same as the first group, got together and wrote another report. And if you're interested in this, the links to the reports are on the Power Point slide.

Well, what did they conclude? The conclusions were slightly different but here's the broad consensus by the experts regarding the importance of IQ tests. And the claim is IQ is strongly related more so — probably more so than any other single measurable human trait to many important educational, occupational, economic, and social outcomes. In some cases, the correlation is very strong such as success in school and success in military training. In other cases, it's moderate but robust such as "social competence." And in other cases it's smaller but consistent, "law abidingness," and they conclude whatever IQ test measure it is of great practical and social importance.

So, IQ matters. More particularly, IQ matters for "social achievement," for "prestigious positions," and for "on the job performance" and other work-related variables. If I know your IQ score, I know something about you that matters. It's not irrelevant just as if I know your score on a personality test of The Big Five I would know something about you that actually would tell me something interesting about you in the real world.

On the other hand, there's a lot of controversy about why this connection exists. So, to some extent, people have worried that the effectiveness of IQ is a self-fulfilling prophecy. And here is why. If society takes IQ tests important — seriously, they become important. So, it's true that your IQ is very related to your success in getting into a good school like Yale. But the reason for this, in large extent, is because to get to Yale they give you an IQ test, the SAT. So, the same for graduate school. There is the GRE, which is yet another IQ test. So, to some extent, it's a self-fulfilling prophecy. I could make — Society could choose to make how tall you are extremely important for educational success. They could say nobody under six feet tall gets into Yale. And then some psych professor would stand up and say, "Of course, height is profoundly related to educational accomplishment," and it would be because people made it so.

So, to some extent, the society that draws highly on IQ tests regarding promotion and educational achievement and military status and so on — it's just going to follow that IQ then becomes important. At the same time, however, the role of IQ is pretty clearly not entirely a social construction. There is some evidence that your IQ score relates to intelligence in an interesting sense including domains like

mental speed and memory span. So, your score on an IQ test, for instance, is to some extent related to how fast you could think and your memory abilities.

Now, I want to shift to the second half of the class and talk about why. So, we talked about two differences, one in "personality", one "intelligence." I want to talk about why people differ but before I do, do people have any questions?

Chapter 4. Question and Answer on Personality and Intelligence [00:30:29]

Yes.

Student: About personality — This morning I took a test — The way the test was, they asked you 100 questions and [inaudible]

Professor Paul Bloom: It's a good question. The question is this young man took — just took a personality test. He was accepted into Slytherin, which is a Hogwarts reference. I'm hip to that [laughter] and — but the question is a good one. You're a clever man, high "G," and you wanted to be in Slytherin. How do we know you didn't work the test? You're going to get these personality tests all the time and the personality tests — You're applying for a business and one of the tests says "I like to steal from my bosses." Well, I don't think so. No. That's a little IQ test right there. So, the question is how do you avoid that problem? The test constructors have done so in certain clever ways. For instance, there are often catch questions designed to catch a liar. Some of these questions pose very unrealistic phenomena so you might have a question in there saying "I have never done anything I am ashamed of." Now, some people will say, "Yes, that's true of me," but they tend to be liars. And so, unrealistic questions tend to catch liars.

Also, you get the same question asked in different ways across the one hundred items and they could use the correlations to figure these things out. Again, the proof is sort of in the pudding. The reliability and validity of a test is determined, in part, by just how well it does at predicting your future performance on the test and your real world performance. And a test that is easily fooled — easily tricked by smart people wouldn't survive long as a personality test. So, we know the test you got is a pretty good test because it seems to work for most people. Yes.

Student: [inaudible]

Professor Paul Bloom: It's a good question. The question was about "Emotional IQ," which is something I'm actually going to touch upon a little bit later in the course, but people have talked about different forms of intelligence. And emotional intelligence, social intelligence, is arguably a candidate for success across different domains. The evidence for its predictive power is not as strong as for regular IQ tests so you might be right. It might turn out to be a much better predictor but one, it's not clear that we know that yet. Peter Salovey has actually done some very interesting research on this and is continuing work along those lines. The second thing is emotional intelligence is actually related to good, old-fashioned intelligence. They kind of pull together a lot. So, it's not entirely separate but that's a good point and I'd like to return to it a little bit later on in the course. Yes.

Student: [inaudible]

Professor Paul Bloom: Good question. How do you determine when — what a good test is? And again, it's a real art going through the details of how to do that but the broad answers involve reliability and validity. It's a good test if I test you today and I test you tomorrow and I get the same score. It's a really good test if your score on that test predicts your grades or, if it's a personality test, predicts how many girlfriends you have or predicts whether people think you're a nice guy. So, you have to see both the replicability of the test over time but also its relationship to real world phenomena. And that's important, again. Why do we know the Batman, Wonder Woman, Hulk test is a bad one? Well, one answer is because what I — how I score on that test isn't going to tell you anything about me. It's not going to relate to my grades. It's not going to relate to how well I'm liked. How do we know the SAT is useful? Well, it actually corresponds with other things like grades. Yes, in back.

Student: [inaudible]

Professor Paul Bloom: Absolutely. The question is — When I'm talking about personality I'm defining it in terms of something which is stable over time. And your question, which is a good one, is, "How do we know it's stable over time?" Can't it change? And the answer is "yeah." A lot of personality does change over time. A personality test you give to a ten-year-old will relate but not so strongly with that individual when he's fifty. On the other hand, we know that the psychological claim that there exists such a thing as personality and it is stable over time. It's supported by the fact that if you're an extrovert now you'll likely be an extrovert twenty years from now. Not perfectly, so you're right. You could change. You could become an introvert, you could become more of an extrovert, but wherever you stand now is significantly related to where you'll be in the future. And that justifies talking about it as a stable trait. Same with IQ. Your IQ might change. It might go up, it might go down, but it won't go up and go down that much and this is why it makes sense to talk about intelligence as a more or less stable trait.

Chapter 5. The Roles of Genes and Environment in Explaining Human Differences [00:36:33]

Okay. Why are we different? Well, you're different because of two things: Your genes and your environment, your nature and your nurture, your heredity and your experience. And this doesn't say anything. This is just defining the question. But the question of the role of genes and the role of environment in explaining human differences is an interesting one and it could be explored in different ways.

But before talking about it I have to clear up a common misconception. I'm going to talk about the effects of genes and I'm going to talk about heredity but I want to be clear. I am talking about the role of genes and also the role of environment in explaining human differences, not in explaining human characteristics. So, the distinction is we're interested in the amount of variation due to genetic differences, not the proportion of an individual's trait that's due to genes.

So for instance, you could pull these apart. The question of — When we ask what's the role of genes, what's the role of heredity in how tall people are, the question is not asking for you — what is the role of your genes in determining how tall you are? It's not clear that's even a sensible question. The question is there's a height difference between you and me and him and her. How do we explain that difference? And I could illustrate why heredity doesn't mean the same thing as the contribution of the

genes. Height is reasonably heritable, meaning the differences between people in the population and how tall they are is due in large part, not entirely, but in large part to their genes.

What about the number of legs people have? Well, the number of legs people have from zero, one or two, is actually not very heritable at all because almost everybody has two legs and people who have fewer than two legs typically have lost one or both legs in an accident. It's not due to their genes. So, of course, whether or not you have legs is a very genetic matter but the differences in number of legs is not usually genetic. And so, heredity is a claim about differences, not a claim about the origin of any specific trait.

Well, now we — That's what heredity, which is genetic — Now, we could talk about environment. And we could break up environment into two sorts of environment. One is shared environment. And shared environment is the extent to which the differences are caused by things — by phenomena that people raised in the same household share. So if one — Suppose some of you are neurotic. And suppose we want to say part of that's due to your environment. Well, suppose you're neurotic because you have lousy parents. That would be part of your shared environment because presumably siblings raised in the same household would have the same lousy parents.

This is contrasted with non-shared environment, which is everything else. Suppose I think you're neurotic because when you were five years old somebody threw a snowball at you and it bounced off your head. That's non-shared environment. Suppose you're neurotic because you won the lottery when you were twenty-one and all the money messed you up. That'd be non-shared environment.

So, what you have here is heredity, shared environment and non-shared environment, and this equals one. That's everything. Non-shared environment is a sort of garbage can category that includes everything that's not heredity and not shared environment. Suppose you think you're neurotic because aliens from the planet Pluto are zapping your brain. Suppose you're right. Well, that would be non-shared environment because they're, presumably, not necessarily zapping your siblings' brains. Everything else is non-shared environment.

It becomes interesting to ask, for all of these differences, the physical differences like height, but psychological differences like personality and intelligence, how do we parcel it out into what's genetic and what's environmental? This proves to be really difficult in the real world because in the real world it's hard to pull apart genes and environments. So, you and me will have different personalities. Why? Well, we were raised by different parents and we have different genes. We can't tell — My brother and me might share all sorts of things in common but we have the same parents and the same genes, fifty percent of the same genes. So how do we tell what's causing us to be alike?

So to do — to pull these things apart you need to be clever. You need to use the tools of behavioral genetics. And to use these tools you have to exploit certain regularities about genes and about environment. One thing is this. Some people are clones. Monozygotic twins are genetic duplicates. They share one hundred percent of the same genes. That's kind of interesting. Dizygotic twins are not clones. They share fifty/fifty. They are just like regular siblings. And adopted siblings have no special genetic overlap. That's zero percent above and beyond randomness. Those three groups then become rather interesting particularly when we keep in mind that by definition two people raised in the same house by the same parents have one hundred percent the same shared environment.

So now, we can start to answer these questions. Suppose you find that monozygotic twins are much more similar than dizygotic twins. Well, that would suggest that there's a large role of genes in those traits that you're interested in. It would not cinch the matter because there are other factors at work. For instance, monozygotic twins look more alike than dizygotic twins and maybe they have different and — they have more similar environments because of this similarity in appearance.

Are monozygotic twins just as similar as dizygotic twins? If so, then it would show that that extra overlap in genes doesn't really matter. And so, it would suggest a low role of heredity. Are adopted children highly similar to their brothers and sisters? If so, then there's a high role of shared environment. Suppose the Bloom children, and there are seven of them, all have an IQ of 104 and we adopt three kids and then at the end of the day those three kids each have an IQ of 104. That would suggest that — And we do this over and over again across different families. That would suggest that there's something about the Bloom family being raised by me that gives you an IQ of 104. On the other hand, if the IQ of the adopted kids had no relationship to those of the biological Bloom children, it would suggest that being raised by me has no effects really on your IQ. It's sort of separate.

A separate — A second — A final contrast, which is the thing that psychologists love, is identical twins reared apart. That's the gold standard because you have these people who are clones but they're raised in different families. And to the extent that they are similar this suggests it's a similarity of their genes. And in fact, one of the most surprising findings in behavioral genetics — The caption here [a cartoon on a slide] is "Separated at Birth, the Mallifert Twins Meet Accidentally." [The cartoon twins] ended up at a patent office with the same device. One of the hugely surprising findings from behavioral genetics is how alike identical twins reared apart are. They seem to have similar attitudes to the death penalty, to religion and to modern music. They have similar rates of behavior in crime, gambling and divorce. They often have been found to have bizarre similarities. They meet after being separated at birth and they meet at age thirty and then it turns out that they both get in to a lot of trouble because they pretend to sneeze in elevators. There was one pair of twins studied by behavioral genetics who were known as the "Giggle Twins" because they were — both would always giggle, they'd burst into giggles at every moment even though it couldn't be environment because they weren't raised together.

More objectively, the brain scans of identical twins reared apart show that their brains are so similar in many cases you can't tell whose brain is who. I could tell your brain from my brain from a brain scan and my brother's brain from my brain from a brain scan. But if I were to have an identical twin it would be very difficult to tell whose brain is whose even if we had no environment in common.

So, this leads to two surprising findings of behavioral genetics. This is the first one. There is high heritability for almost everything. For intelligence, for personality, for how happy you are, for how religious you are, for your political orientation, there — for your sexual orientation, there is high heritability. There's a high effect of genes for just about everything.

Now, that's actually not the controversial thing I'm going to tell you. But before getting to the more controversial thing I want to raise another issue which often gets discussed and has a good treatment in the textbook. This suggests that individual differences within this — within a group have genetic causes. Does that mean that group differences are largely the result of genetic causes? So, we know that there are clear differences in IQ scores among American racial groups, between whites and Asians,

African Americans, Ashkenazi Jews. There's clear and reliable IQ differences as well as some other differences.

Now, to some extent, these groups are partially socially constructed. And what this means is that whether or not you fall into a group it's not entirely determined by your genetic makeup. It's often determined by social decisions. So, whether or not you count as a Jew, for instance, depends not entirely on genetic factors but also on factors such as whether you're reform or orthodox and whether you — so whether you would accept that a child of a Jewish man and a non-Jewish woman is Jewish. Similarly, categories like African American and white and Asian often overlap broad genetic categories and they don't make fully coherent genetic sense.

At the same time though, there is plainly some genetic differences across human groups and say with regard to vulnerability to disease. Ashkenazi Jews for instance are vulnerable to Tay-Sachs. And the fact that you could have this sort of genetic vulnerability suggests that there is some sort of reality to these groups. So, you have to ask the question now, to what extent does the high heritability in individuals mean that there has to be a heritable explanation across groups? And the answer is "not at all." I'm not saying that this means that there's no genetic explanation for human group differences. All I'm saying is the question of the phenomena of — within-group genetic differences does not mean that there is across-group genetic — sorry, between-group genetic differences.

There is a nice example by Richard Lewontin, the geneticist, where he imagines two plots of — what are you — some sort of wheat, yeah, two plots of land and each one has a set of seeds and — Oh, no. They're over there. [pointing to slide] No. Anyway, one of them you fertilize a lot. The other one you fertilize a little. Now, within each plot how much the seed grows is actually largely determined by the genetics of the seed. And so, you'd find high heritability for growth in the seeds. But the difference between groups has no genetic cause at all. It's caused by which groups you fertilize more than others.

Here's another way to do the logic. Suppose from the middle [aisle of the classroom] down here, you guys, [pointing to the people on the right] I hate you, I really hate all of you, and [pointing to the people on the left] I like you, so I make up two Midterms. You probably didn't notice but there were two Midterms. This Midterm was fiercely hard, savagely hard [pointing to the people on the right]. It took many of you until the end of class to do it. This Midterm [to the people on the left] was, "Which is bigger, a dog or an elephant?" [laughter] because I like you and I want you all to succeed.

So, you have two different groups, you guys and you guys. Within each group some people are going to do better than others. The explanation for that might actually have to do with your genes. It might have to do with your environment, how much you study, but all sorts of reasons for that. Within each group some of you will do better on the hard test than others on the hard test, some better on the easy test than others on the easy test. But how do we explain the group difference? Well, it has nothing to do with genes. The group difference, the fact that you will do much worse than you, has to do with the exams I give. My point, again, is that there is a logical difference between a within-group difference, within this half of the class, and a difference between groups, within — between this group and this group.

Chapter 6. Genes, Environment and Intelligence [00:51:16]

What do we know about — So, that just shows they're not the same thing but what's the fact of the matter? What do we know about human differences between different human groups? Again, the textbook has a good discussion of this but I'm going to give two reasons from the textbook that at least group differences are at least to a large extent due to environmental and not genetic causes. One is that the differences we find in IQ seem to correspond better to socially defined groups than genetically defined groups. They seem to correspond to groups defined in terms of how people treat you and how people think about you as opposed to your DNA. And to the extent that turns out to be true that would mean that a genetic explanation is not reasonable for those differences.

A second factor is that we know IQ can differ radically without any genetic differences at all. And the most dramatic evidence of that is the Flynn effect. The Flynn effect is one of the freakier findings. The Flynn effect is the finding that people have been getting smarter. You are much smarter on average than your parents if — and the IQ tests hide that. Here is why they hide that. They hide that because they always make 100 the average. So, you come home and you say, "Dad, Dad, I just did an IQ test. I got 120." And your father says, "Good work, Son. I got 122 when I was your age," but what neither of you acknowledge is your test was much harder. As people got better, they had to make the test harder and harder. And this is plotted by the Flynn effect.

[referring to a graph] One of these lines is American and one is Dutch. I don't know which is which but the gist of it is that somebody who would have — that if you in 1980 would take the 1950 test, your average person in 1980 would score 120 on the 1950 test. What this means is if you take your person who's average now and push him back through time twenty years, thirty years, he would do much better than average. Nobody knows why people are getting smarter and there's different theories of this. And in fact, well, wait until you see your reading response. But what this illustrates is that IQ can change dramatically over the span of a few decades without any corresponding genetic change. And that leaves open the possibility, in fact, maybe the likelihood, that the differences we find in human groups, existing human groups, are caused by the same environmental effects that have led to the Flynn effect.

Okay. This is not the surprising claim though, the high heritability for almost everything. This is the surprising claim. Almost everything that's not genetic is due to non-shared environments. The behavioral genetic analyses suggest that shared environment counts for little or nothing. When it comes to personality or intelligence then, an adopted child is no more similar to his siblings than he or she is to a stranger. To put it a different way, the IQ correlation in genetically unrelated adults who are raised in the same family is about zero. Suppose the Bloom family all has an IQ of 104 and we adopt a kid. What will this kid's — We adopt him as a baby. We raise him to be a twenty-year-old. What's his IQ? Answer? We have no idea because the IQ of the Bloom family who are unrelated to him has no effect at all.

Now, if you think about the implications of it, it becomes controversial and Newsweek, I think, caught the big issue when they put in their title the question "do parents matter?" And the question — And the issue is parents are shared environment. To say shared environment does not affect your intelligence or your personality suggests that how your parents raised you does not affect your gene — your intelligence or your personality. This isn't to say your parents didn't have a big effect on your

intelligence and personality. Your parents had a huge effect on your intelligence and your personality, around 0.5 actually. They had this effect at the moment of conception. From then on in, they played very little role in shaping you — what you are.

The case for this which generated the Newsweek cover came up in a controversial book by Judith Harris called The Nurture Assumption which has a very long subtitle, "Why Parents Turn — Why Children Turn Out the Way They Do, Parents Matter Less than You Think and Peers Matter More." Judith Harris has had an interesting history. She was kicked out of graduate school at Harvard and told that she wouldn't amount to much. The person who wrote the letter saying that she was not going to amount to much was the department chair, George Miller. In 1997, she won the George Miller award for her astounding accomplishments. And when she wrote the book she took as a starting point, her point of disagreement, a famous poem by the poet Philip Larkin and many of you have probably heard this. The poem goes like this:

They fuck you up, your mum and dad.They may not mean to but they do.They fill you with the faults they hadAnd add some extra just for you.

The last line of the poem, the last bit of the poem, ends: "Man hands on misery to man. It deepens like a coastal shelf. Get out as early as you can and don't have any kids yourself." It's beautiful. [laughter] Harris wrote a rebuttal: "How sharper than a serpent's tooth to hear your child make such a fuss. It isn't fair. It's not the truth. He's fucked up, yes, but not by us." [laughter]

Just to show that academic debates never end, a British psychoanalyst named Oliver James, outraged by Judith Harris' book The Nurture Assumption, wrote another book in response called They Fuck You Up. [laughter] Now, how do you tell your grandparents, "I wrote a book." "What's it called?" "Can't tell you." [laughter]

Anyway, look. If you're paying attention, this has to sound wrong. You must be thinking of course there must be an effect of shared environment. Of course parents have an effect. After all, good kids have good parents. There is no doubt at all that this is true. There is a high correlation between parent and child for everything. If your parents read a lot and there's a lot of books in your house, you will become a reader. If your parents are religious, you will be religious. If you're raised by Bonnie and Clyde, you will be a young thug. [laughter] If your parents are poor, you're likely to be poor. If your parents are brilliant, you're likely to be brilliant. No doubt at all. It is an extremely robust correlation. But the problem is this correlation could be explained in different ways. Everybody thinks it's because parents do something that affects their kids. Your parents are bookish, they read to their kids, so their kids become bookish, but another possibility, which we know is true, that almost always parents share their genes with their kids.

Another possibility is it's the parents who are affecting — sorry, it's the child who is affecting the parents, not vice versa, and to illustrate this, these different possibilities, I want to tell you a little bit about a study. And I really find this a fascinating study. It was reported last year and it was a study shown that — suggesting that family meals help teens avoid smoking, alcohol, drugs. It involved a phone questionnaire where they phoned up teenagers and their parents and said, "Hey, teenager. Do you do a lot of drugs?" "Yes." "Do you have dinner with your parents?" "No." And they take it off — and

then they ask other people and they find that the kids who are the good kids have meals with their parents, suggesting this headline.

I like this study because I have read — I must have read in my career a thousand studies and this is the worst study ever done [laughter] in the history of science. And it's almost — We could devote a week to discussing what's wrong with this study. Let's just — But here's the idea. It is possible that they are right. It is actually possible — there's no — I have no evidence against it – that having meals with your kids makes them into good, drug-free, non-promiscuous, non-drinking kids. Of course, it's equally possible it's the other way around. If little Johnny is kind of — is out there smoking pot and cavorting with prostitutes and stuff like that, he's not going to come home for the family meal. It's the other way around. While if he's a good kid, he might be more likely to have a family meal. So, the direction — It might actually be not family meals make good kids but rather good kids stick around to have — if they have nothing better to do and have meals with Mom and Dad. [laughter]

Another possibility is there's good families and bad families. A good family is likely to have drug-free kids and a family meal. A bad family is likely to have stoned kids and no family meal. [laughter] So, there — maybe there's an effect of that. The parents had nothing to do with the family meal.

Here's the even weirder part. They didn't factor out age so think about this. Their sample included children ranging from twelve to seventeen but let me tell you something about twelve-year-olds. Twelve-year-olds don't use a lot of drugs and are likely to eat with their family. Seventeen-year-olds are stoned all the time and they don't eat with their family. [laughter] I've just begun on this study but the point is when you hear something like — So now, take something which you may be more likely to believe. Maybe you believe that having parents who read to their kids, that's good for their kids. Well, maybe it is but most of these criticisms apply to that study too. A bookish kid is more likely to get his parents to read to him. A good family — Parents who are good parents in general are more likely to do all sorts of good things to their kids and have good kids besides.

Take another case, the so-called cycle of violence. Yes, it's true. Parents who smack their kids tend to have statistically more violent kids. But maybe the causality goes the other way around. Maybe if you have a kid who is a troublemaker you're more likely to smack him. Maybe, which seems to be entirely likely, the propensity for violence is to some extent heritable. And so, even if the kid was not raised by the smacking parent, whatever properties of that parent caused him — led to that violence got inherited by the kid.

Now, again, this isn't going to sit right for you and I've had — I put this down because last year when I gave this talk people ran up to me and told me this. They said, "Look. I know my mom and dad had a huge role in my life. That's why I'm so happy and successful," then other people said, "That's why I'm so miserable and screwed up," but either way blame it on Mom and Dad or thank Mom and Dad. And you might think you know. When you become famous and you stand up and you get your awards maybe you'll thank your mom and dad. When you go to your therapist and explain why you're so screwed up maybe you'll blame Dad. "He never took me to a baseball game." Well, maybe, [laughter] but you don't know. Were you adopted? If you weren't adopted, you can't even begin to have the conversation about how your parents messed you up because if you're a lot like your parents you might be a lot like your parents because you share their genes. Of course, you resemble your parents. Moreover, how do you figure out which is the cause and which is the effect? "Mom smacked me a lot

and that's why I turned out to be such a rotten person." Well, maybe she smacked you because you were rotten. [laughter] I don't want to get personal but it's very difficult to pull these things apart.

A final point on this. One response to Harris' book is this. "Look. Even if this is true, you shouldn't let this get out because if parents don't mold their children's personalities maybe why should they treat their kids nicely?" And you might be wondering this. You might be thinking, well, gee, if you don't have any effect on how your kids turn out, why be nice to them, but there are answers. You might want to be nice to them because you love them. You might want to be nice to them because you want them to be happy. You might want to be nice to them because you want to have good relationships with them. And I have a little bit more but I'm going to skip it and I'm going to move right to your reading response, which is very, very simple, easy to answer, easy to grade: Explain the Flynn effect. It's a toughie so just explain that. Okay. Have a wonderful spring break and I'll see you when you get back.